Aryan Wadhwa

+91 - 8178156168 · aryanwadhwa911@gmail.com · LinkedIn · GitHub · Medium

SKILLS

- **Programming Languages:** C/C++, JavaScript, TypeScript, Python
- **Frontend Development:** HTML, CSS, React.js, Next.js, Tailwind CSS
- **Backend Development:** Node.js, Express.js, Mongoose, Prisma
- **Databases:** MongoDB, MySQL, PostgreSQL
- **AI / ML:** Scikit-learn, PyTorch, TensorFlow, Keras, Pandas, NumPy, HuggingFace, LangChain, LangGraph
- **Cloud & Tools :** AWS, Docker, GCP, Git, GitHub

EXPERIENCE

Software Engineer Intern – Greenleap Robotics (**IIT Delhi** Incubated), New Delhi, India **Sept 2026 - Nov 2026**

- Architected a scalable, distributed IIoT platform from scratch, designing device-to-cloud architecture to orchestrate **1,000+** autonomous robots across **55+** solar plants (**2+ GWp**).
- Designed and developed **5+ backend microservices**, **20+ REST APIs**, and **7 real-time event** channels for authentication (**4-tier RBAC**), fleet visualization, robot telemetry, and remote mission control; integrated **STM32** firmware via **MQTT/TLS** and WebSockets, achieving sub-50ms **p95 latency**.
- Engineered high-throughput, event-driven pipelines with direct **MQTT** to **WebSocket** streaming, hybrid caching, and asynchronous **batch processing, database latency** by **85%** and deployed containerized services on **AWS EC2/S3** using Docker.

Quantitative Researcher Intern – Kosmoz Consulting**,** San Francisco, CA (remote) **Jan 2026 - May 2026**

- Engineered an automated algorithmic trading engine covering **45+ US equities** and **ETFs**, integrating backtesting, forward testing, execution modeling, and **real-time alerts**.
- Developed **5+ multi-factor trading strategies** combining EMA crossovers, momentum, volatility regimes, and **Markov Chains** to automate market analysis and **trade signal generation**.
- Generated **15%+** portfolio returns in **1 month** of **forward testing** & **algo based** trading across US tech equities, ETFs, and Granite shares through systematic strategy execution.
- Engineered **dynamic position sizing** and portfolio **risk optimization**, evaluating **6+ performance metrics** including Sharpe, Sortino, Alpha, Beta, CAGR, and Maximum Drawdown for capital allocation and risk-adjusted returns.

AI/ML Researcher Intern – PavePilot AI (**YC - W24**), San Francisco, CA (remote) **Jul 2025 – Aug 2025**

- Programmed **AI automation pipelines** for multi-platform content deployment through business APIs, increased API performance by **60%** and automated core **operational workflows**, reducing **task execution** time by **40%**.
- Developed AI-powered **LangChain** workflows, driving **65% higher** user engagement via autonomous processing of **10,000+ daily multi-modal queries.**
- Optimized core **Backend services** scalability by integrating **GCP** and **automating CI/CD** for faster, reliable deployments.

PROJECTS

Deep Packet Analyzer (DPI Engine): GitHub / blog

Tech Stack: C++, PCAP, TCP/IP, TLS, Multi-threading, CMake, Concurrent Queues

- Built a high-performance **Deep Packet Inspection** (DPI) engine in C++ to parse raw PCAP traffic with Ethernet/IP/TCP decoding and **TLS** ClientHello **SNI** extraction.
- Engineered a scalable multi-threaded architecture using **mutexes**, **condition variables**, thread-safe queues.
- Implemented 5-tuple flow tracking and rule-based filtering for **traffic classification** and **blocking**.

Stock Prediction & Analysis Platform (StockAI): GitHub / web-link

Tech Stack: Next.js, TypeScript, Tailwind CSS, TensorFlow, Recharts, WebSockets, REST APIs, LangChain, Transformers

- Built a **full-stack** web app for real-time stock price prediction using **LSTM models** and integrated technical indicators (RSI, MACD, Bollinger Bands).
- Implemented an **AI-powered LLM system** using **LangChain** and **Transformers** for portfolio insights and market analysis.
- Deployed full-stack platform with **CI/CD** pipelines using **AWS lambda**, optimized for performance, scalability, and security, attracting 100+ **active users** for stock market insights.

Bolt2.0 (AI- powered Website Maker) : GitHub

Tech Stack : Next.js, Tailwind CSS, Convex DB, Prisma, Grok API, GCP.

- Developed an **AI-powered** platform that transforms **natural language** prompts into production-ready **full-stack Next.js** applications with **responsive UI.**
- Attracted 500+ users & Implemented **real-time preview**, collaborative editing, and **automated deployment**, reducing development time by **95%** using GCP.

ACHIEVEMENTS

1. Delivered ML solutions adopted by **Harvard Incubation Center**, contributing to heart disease prevention and Risk analysis in **first internship**.
2. Hackathon **Mentor & Judge**: Mentored **70+ teams** at SuperNova'25 and judged HackQuanta, evaluating AI/ML and Full-Stack projects.
3. **OSCI Mentor** – Open Source Contribution India (Aug 2025).
4. **Hackathon Leadership Highlights:** Secured **3rd position** in T`Hacks (organised by IEEE club) & Top 5 in **10+** hackathons with unique problem statements building & **SIH** university level qualified (2024 & 2025).
5. Active member, GDG Club, **AI/ML** & **UI/UX** (Nov 2024 – Jun 2025).
6. Head Boy, representing **1,500+ students**; led student council, coordinated events, and contributed to policy enhancements.

Research & Publications

Addressing Memory Limitations in **LLM-Based AI Agents** : An Architectural Evaluation of Modern Memory Management Frameworks.

Authors : **Aryan Wadhwa**, **Astitva Gaur**

Publication Links : DOI:10.5281/zenodo.19483083 | ResearchGate Link (Nov 2025 - May 2026)

- Conducted a systematic evaluation of memory architectures in large language model (LLM) agents, benchmarking systems like RAG, MemGPT, HiAgent, LangGraph, LangChain, A-Mem and Vectorless DB across scalability, latency, and long-term reasoning.
- Implemented high end case scenarios for memory limitation and created embeddings for semantic chunk retrieval and its latency from the given AI agents. GitHub

EDUCATION

B.Tech. Computer Science and Engineering (Sept 2023 - June 2027)

Guru Gobind Singh Indraprastha University | **CGPA : 9.4/10,** Expected 2027

J.D Tytler School (2010 - 2023)

- 12th Class : **93%** (PCM) in 2023
- 10th Class : **91%** in 2021